Aman Verma

AI/ML ENGINEERING INTERN

Jaunpur, UP, India | +91 98765 43210 | amanverma.dev@gmail.com | linkedin.com/in/amanverma-765 | github.com/amanverma-765

TECHNICAL SKILLS

Languages & Core: Python, C++, SQL, Bash

Cloud & Infrastructure: Docker, Linux, Git

Databases & Storage: PostgreSQL, Redis, SQLite (FTS5), ChromaDB

Frameworks & Developer Tools: PyTorch, FastAPI, Celery, Pydantic AI, Hugging Face, pytest

WORK EXPERIENCE

Software Engineering Intern – **LexoLegal** June 2024 -- Aug. 2024
Remote

- Architected multi-tenant legal search engine in **Python** and **FastAPI** using **SQLite FTS5**, reducing query latency from **850ms to 110ms** across **40,000 regulatory documents**.
- Engineered asynchronous worker queue with **Redis** distributed caching and **Celery**, processing **15,000 document transformations daily** under **Linux**.

PROJECTS & OPEN SOURCE

ncert-rag-evals | *Python, PyTorch, ChromaDB, Hugging Face, Docker, pytest* | https://github.com/amanverma-765/ncert-rag-evals Sept. 2024 -- Present

- Architected automated evaluation framework across **84 test suites** via **Git** and **Bash** using **pytest**, achieving **350 evals/sec** throughput (**p < 0.01**).
- Engineered semantic retrieval pipeline in **ChromaDB** with **Hugging Face** and **PyTorch**, reducing embedding retrieval latency by **42%** across **12,000 chunks** using **Docker**.

doubtless | *Python, Pydantic AI, PostgreSQL, SQLite, C++* | https://github.com/amanverma-765/doubtless Jan. 2024 -- May 2024

- Developed local-first hybrid retrieval engine in **Python** and **C++** bindings fusing BM25 with dense embeddings in **PostgreSQL**, achieving **86.9% Recall@5 in 8.9ms on CPU**.
- Secured multi-tenant runtime isolation by binding tenant context inside **Pydantic AI** dependencies, preventing prompt-injection state leakage across sessions with **SQL** constraints.

EDUCATION

Indian Institute of Information Technology 2021 -- 2025
Bachelor of Technology in Computer Science and Engineering *India*

Cumulative GPA: 8.6 / 10.0